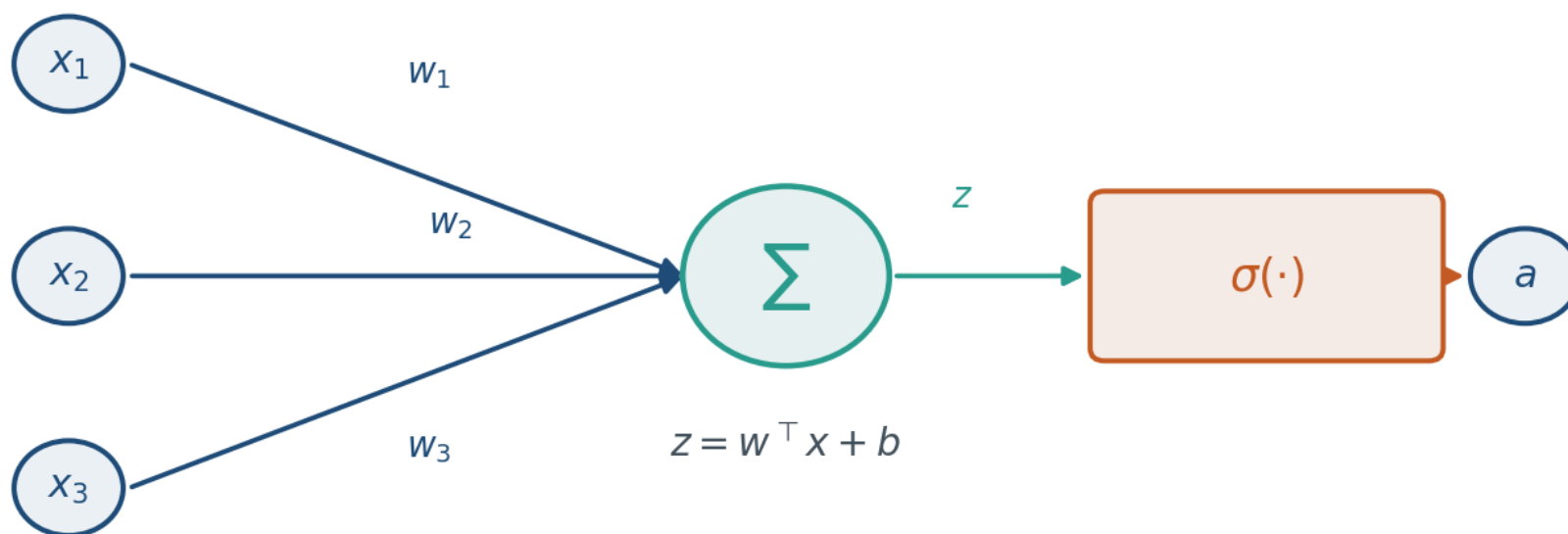


One neuron

- Pre-activation $z = w^\top x + b$, then $a = \sigma(z)$.
- w and b are the parameters you train.
- ReLU in hidden layers; softmax when classes compete.

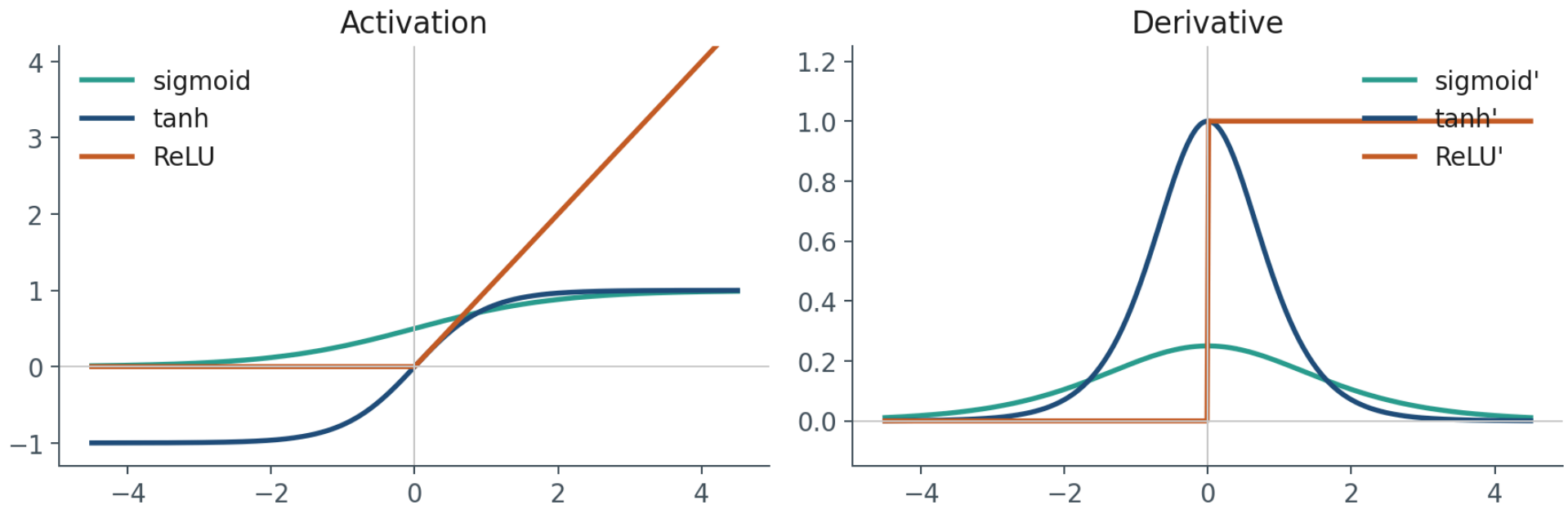
One neuron: weighted sum, then a nonlinearity



The nonlinearity is the point

- Two affine maps compose to one affine map.
- Without σ , depth does not add power.
- Sigmoid saturates. ReLU is cheap and sparse.

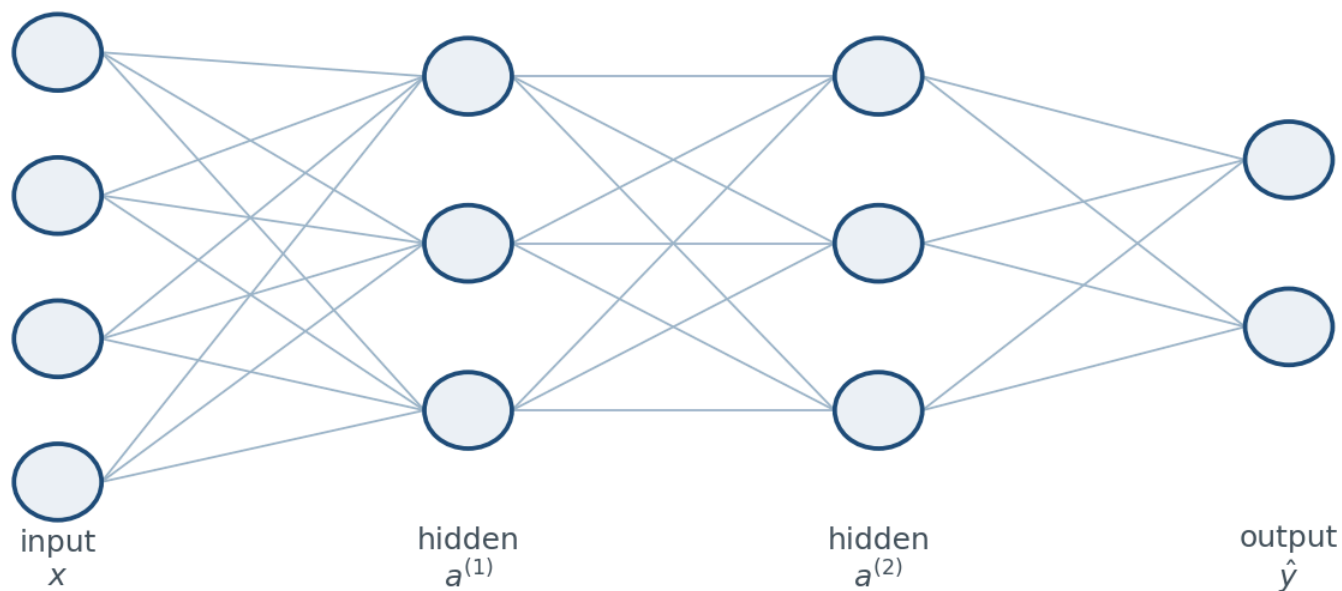
Without a nonlinearity, stacked layers collapse to one linear map



A feedforward net

- Layer ℓ : $a^{(\ell)} = \sigma(W^{(\ell)}a^{(\ell-1)} + b^{(\ell)})$.
- $a^{(0)} = x$. Every arrow is one weight.

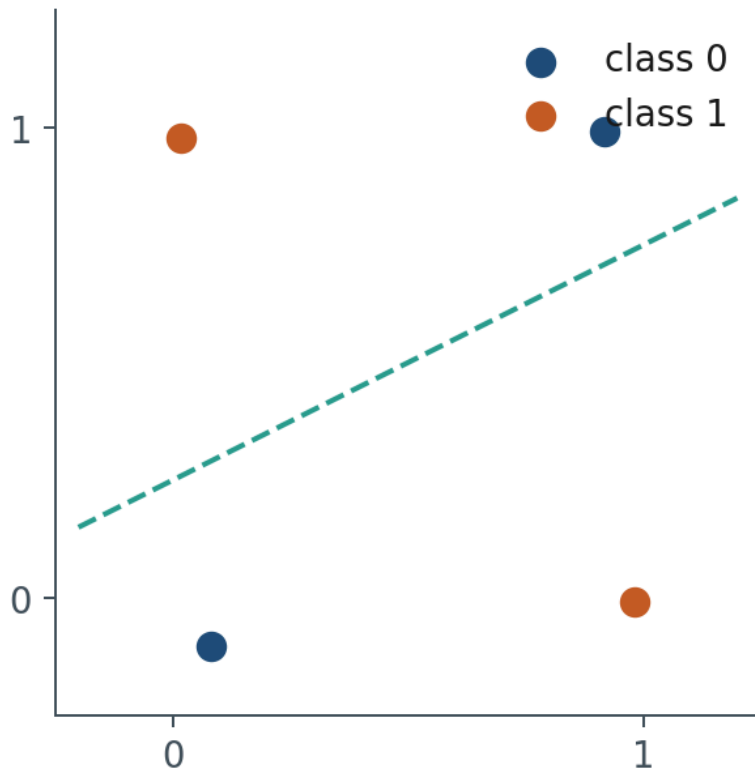
A feedforward net: each layer is $a^\ell = \sigma(W^\ell a^{\ell-1} + b^\ell)$



XOR: why a hidden layer exists

- One linear unit cannot separate XOR.
- A small MLP can fold the space so the classes split.
- Language models stack many such nonlinear maps.

One linear cut cannot do XOR



A hidden layer can bend the boundary

